

on C, C depends on A.

In the simplest of cases, the prerequisites are all files. But there may be situations in which you would like to use a directory as one of the prerequisites. *make* enables you a way to handle this too.

Imagine the following scenario: you have a directory *lff/* where you have files *lff1*, *lff2*, *lff3*. You usually zip all the files of *lff/* and keep it as a fallback copy. Every time a new file is added to *lff/*, or when any of the existing files in *lff/* is updated, you would like to zip *lff/* again, and store all files in a single zip file *lff.zip*.

You can write a simple rule with the target as *lff.zip*, and prerequisites as *lff1*, *lff2*, *lff3*. But if you have 1000 files in *lff/*, you cannot obviously list each of them by name. Or, if files keep getting added or removed from *lff/* frequently, you don't want to keep rewriting the rules each time. You would like to use a wildcard expansion of *lff/* and let the wildcard do all the book-keeping. Try the following experiment:

1. Create a directory *lff/* and add the dummy files *lff1*, *lff2* and *lff3* in it using touch. In fact, you can fill up the directory with a 1000 files, if you want.
2. Create a blank file *testmake* and enter the following in it:

```
# testmake start
# Calling sequence "make -f testmake"
objects := $(wildcard lff/*)
lff.zip : $(objects)
    echo lff is more recent
    zip lff.zip lff/*
#end testmake
```

3. Run *make*, using *make -f testmake*. It will create *lff.zip*.
4. Run *make* using *make -f testmake*. Nothing happens, since *lff.zip* is up-to-date.
5. Now run touch *lff/lff1* and run *make -f testmake*. It will update *lff.zip*. In fact, *lff.zip* will get updated every time any file in the *lff/* directory is touched (updated).
6. Remember, *lff.zip* should not be in the *lff/* directory, since it will cause an endless loop in the above script.

You can get all the files of the above experiment along with documentation, as a single zip/tar bundle by sending a mail to drpartha@gmail.com. In fact, this is a more elegant solution than the one I proposed in an earlier article in *LFY* (*Treasure Hunt with Find*, July 2008). 

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